Dynamics of the Multi-Stock Artificial Market Consisting of Risk-Seekers and Risk-Avoiders

Hidenori MABE † Yu CHEN ‡ Fujio TORIUMI † and Hirotada OHASHI †

† Department of Systems Innovation, the University of Tokyo 7-3-1 Hongo, Bunkyo-ku, Tokyo, 113-8656, Japan

‡ Department of Human and Engineered Environmental Studies, the University of Tokyo 5-1-5 Kashiwanoha, Kashiwa, Chiba,277-8561, Japan

E-mail: † mabe@crimson.q.t.u-tokyo.ac.jp

Abstract

We construct a multi-stock artificial market model composed of two types of traders, one type corresponds to risk-avoiders who tend to invest in bargain-priced stocks and the other to risk-seekers who tend to invest in highly volatile stocks. The dynamics of this model becomes complex because of the interactions between traders' behavior and price changes of stocks. Some stylized facts such as fat-tail distribution and volatility clustering are observed by this model. We investigate the characteristics of the model comparing with those of real markets. Especially, we focus on "volatility anomaly", one of the peculiar phenomena observed in real markets.

Keyword: agent-based model, artificial market, stylized fact, volatility anomaly

1. Background

In artificial stock market study, the most of models were based on a single brand and focused on its dynamics, however, there is no independent market of a single brand in reality. Considering artificial markets including several brands of stocks is important because actual traders do not fix what to buy, but always switch their positions among different brands. The complexity of the interactions is partly caused by the actions of various traders, who respond differently to the market behavior. In this study, we construct an artificial market model of several brands of stocks in which agents switch their positions.

2. Model

We constructed an agent-based model of a stock market comprised of five brands. This model also includes a random walking time series (r_t) representing an exchange rate in order to consider external impacts against fundamentals of stocks ($EV_{i,t}$).

$$EV_{i,t} = EV_{i,t-1} + sensitivity_i * r_t$$

Related with position switching actions, we introduced two types of traders, risk-seekers and risk-avoiders. A strategy of each trader type is as follows.

Risk-seeker:

$$Position_{j,t} = arg\ max\ (historical\ volatility_{i,j,t})$$

Risk-avoider:

$$Position_{j,t} = arg\ max(EV_{i,t}/Price_{i,t-1} * (1 + intuition_j * random))$$

A price of each stock is determined by these agents' positions, and then, agents decide their next positions based on the prices.

3. Results

One of the results of calculation is shown in Figure 1. In this case, the number of agents is N=2000, the ratio of risk-seeker to risk-avoiders is 1, and the number of stock brands is 5. Agents can take only long-positions. As this figure shows, prices do not follow the exchange rate directly, instead, change complexly. This result reflects the interactions between two types of traders. The series of price returns indicate some stylized facts, such as fat-tail distributions, volatility clustering and auto-correlations for short lags.

4. Conclusion

We simulated complex behavior of multi-stock markets including two kinds of traders. Some stylized facts were observed in this simple model. "Volatility-anomaly", which is a phenomenon contradicting the modern portfolio theory, could be reproduced under some situations in this model. The mechanism of this phenomenon is under investigation.

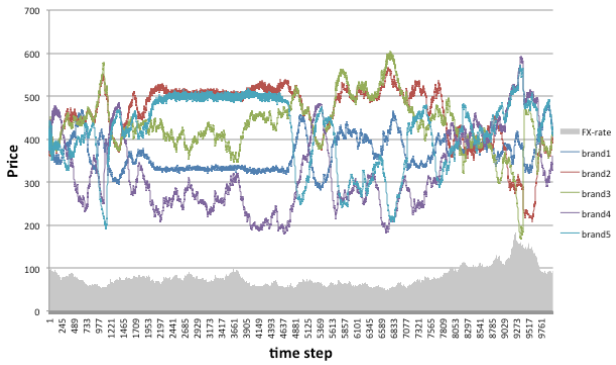


Figure 1. Stock prices and exchange rate